

Ankit

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Education

Indian Institute of Technology (IIT), Ropar

Bachelor of Technology, Mathematics and Computing

2020 – 2024

CGPA: 7.7/10

Experience

Futures First, Bangalore

Aug 2024 – Present

Analyst

- Traded SOFR and Fed Funds futures (SR3, ZQ) across outright, calendar spreads and butterflies, analyzing CPI, NFP, GDP and FOMC releases for rate path implications and positioning across the short-term interest rate curve.
- Configured automated execution algorithms (RangeTrader, ASE, ECI) for range bounded order management, multi-leg quote coordination and macro event triggers, reducing execution latency during high volatility releases and ensuring deterministic fill logic.
- Independently built and deployed a cross asset macro dashboard (FastAPI + React, Render/Vercel) ingesting live FRED and Yahoo Finance data to visualize SOFR curve shifts, energy and FX market reactions, and AI-generated analyst commentary around macro events, replacing manual post event analysis across multiple trading terminals.

Samsung, Bangalore

Jun 2023 – Jul 2023

SDE Intern

Certificate

- Developed a semi-supervised **anomaly detection** pipeline for large scale system log data using **LSTM** based deep learning, reducing false positives on unlabeled data through self-supervised pretraining.
- Designed and implemented four **RESTful API** endpoints for session lifecycle management (create, save, retrieve, list) in an internal developer tool improving backend modularity and reducing session-handling bugs reported by the team.

Projects

Macroeconomic Event Impact Analyzer and Cross Asset Intelligence Dashboard

Oct 2025 – Present

Python, FastAPI, React, SQLite, FRED API, Google Gemini API

GitHub Live-Demo

- Built and deployed a full-stack market analytics platform (**FastAPI**, **React**, and **SQLite**) with 10+ REST API endpoints and an automated data pipeline ingesting live macroeconomic and market data from FRED and Yahoo Finance across rates, energy and FX, hosted on Render and Vercel with CI/CD via GitHub.
- Developed a cross asset macro reaction dashboard tracking simultaneous movements across rates, energy and FX markets around CPI, NFP, GDP and FOMC releases with SOFR/ZQ rate structure builder
- Integrated **Google Gemini API** with custom prompt engineering to auto-generate structured macro analysis (hawkish/dovish classification, rate path commentary) on each data release, reducing manual interpretation from 15 mins to under 30 seconds per event.

Credit Risk Analysis

Jan 2024 – May 2024

Python, Pandas, NumPy, Scikit-learn

GitHub

- Built a Python data pipeline to analyze company financial data and predict credit risk using **Logistic Regression**, **Random Forest**, **SVM**, **CNN** and **LSTM**; identified MScore as the most important feature and achieved 84% accuracy.
- Implemented portfolio optimization using **Markowitz Mean Variance** and **Black-Litterman Allocation** models and developed a web interface to visualize risk, returns tradeoffs and optimized portfolio allocation.

Real-Time Face Recognition System

Mar 2023 – Apr 2023

Python, HTML/CSS, OpenCV, CNN – Guided by Dr. GS Raju, IIT Ropar

GitHub

- Developed a web-based face recognition system for college attendance and gate entry/exit by integrating a live camera feed with a Python backend and **OpenCV**-based face detection pipeline.
- Implemented real-time face preprocessing and **CNN**-based recognition (multi-face detection, resizing, normalization, grayscale optimization) to generate embeddings per frame and automate attendance logging.

Technical Skills

- Programming Languages:** C/C++, Python, HTML/CSS, SQL
- Libraries and Tools:** Git, PostgreSQL, NumPy, Pandas, Matplotlib, REST APIs, Scikit-learn, TT, ADL
- Frameworks and Platforms:** FastAPI, React.js, OpenCV, Render, Vercel, SQLite

Key Courses

- Computer Science:** Data Structures, Algorithms, Object Oriented Programming, Databases and Management Systems
- Mathematics:** Probability and Statistics, Data Science, Mathematical Image Processing, Calculus and Linear Algebra

Achievements

Top 5% – Google CodeJam, Google KickStart and Facebook Hacker Cup,

2022

Institute Merit Scholar, (IIT Ropar)

2021

National top 0.3%, among 1.2M candidates in JEE Mains

2020

Competitive Programming, Solved 600+ problems on LeetCode; rated on Codeforces & CodeChef

Profile

1st Position, Maths Quiz Cynosure, National Symposium on Advances in Mathematics

Certificate

Tech Fest Head, Led a team of 50, Organized Technical workshops on AI/ML & AR-VR, Lecture Talks, Army Exhibitions, providing technical solutions for problems faced by Indian Army.

2023